Sequence

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Learning Journey and Evidence

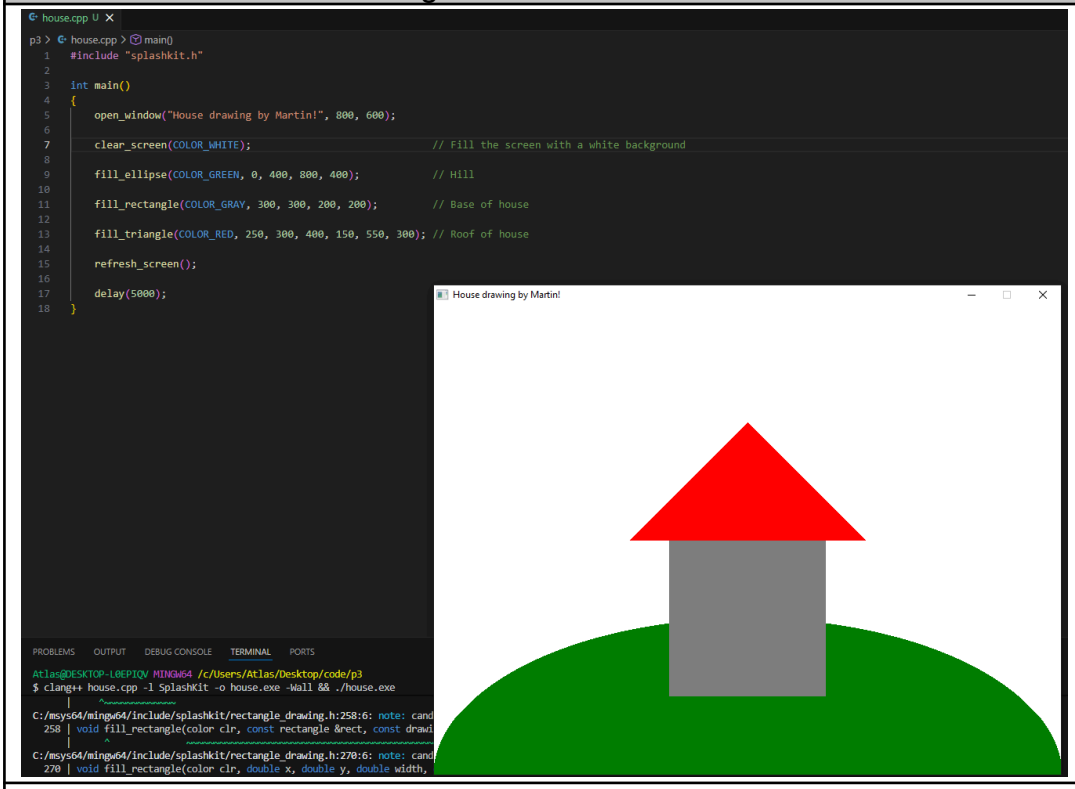
Prior Knowledge and Experience

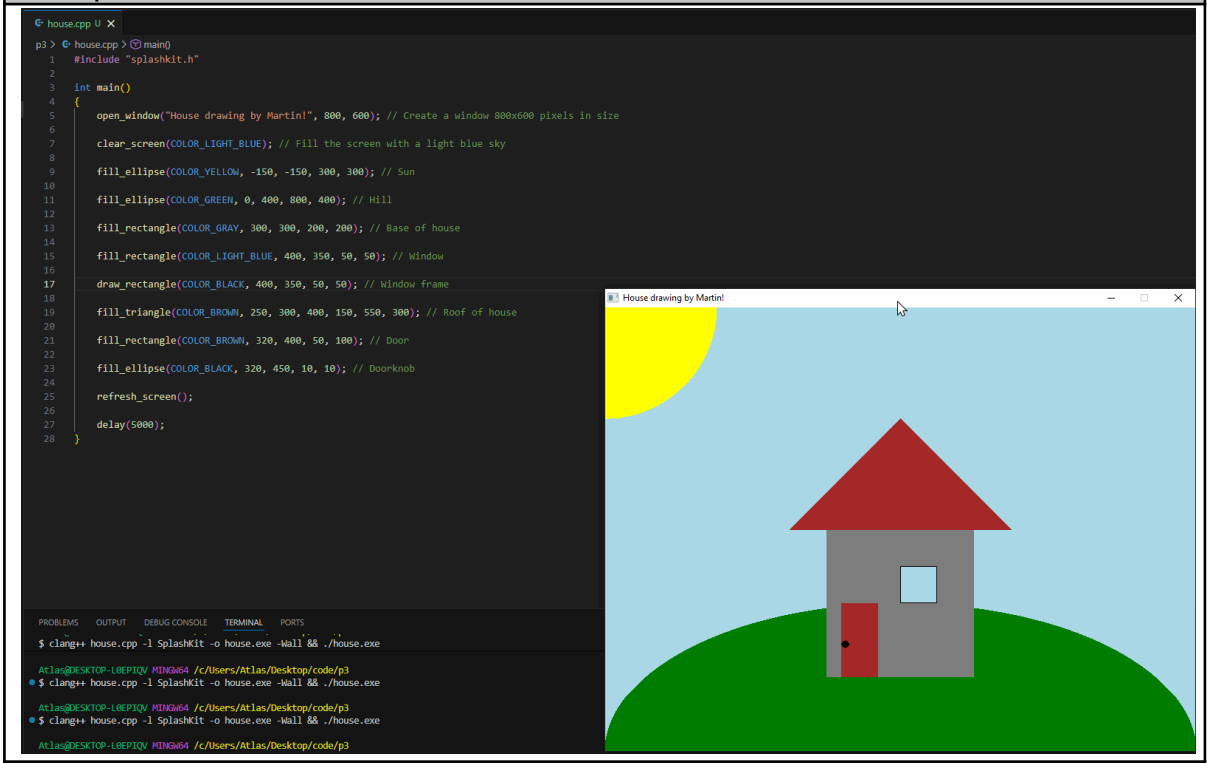
Overview Summary

A computer executes programs in a **sequence**, meaning one instruction at a time is carried out. A **program counter** keeps track of the instruction currently being executed, and is incremented by 1 to indicate execution of the next instruction in the sequence.

Study to Acquire, Refresh, and/or Extend Capability

To study and master this topic I did the following:

Guided tour – House Drawing

I'm not too familiar with the C++ syntax e.g. I'm not used to putting semicolons after every line. However, I managed to replicate the house from the guide.

In Depth - Reference

I added some additional elements to the house, including a subtle frame for the window using the draw_rectangle function from the graphics procedures reference. Finding the correct coordinates was quite difficult and took a lot of trial and error as can be seen by the repeated compilations.

```
house.cpp U
p3 > house.cpp > main()
#include "splashkit.h"

int main()
{
    open_window("House drawing by Martin!", 800, 600); // Create a window 800x600 pixels in size

    clear_screen(COLOR_LIGHT_BLUE); // Fill the screen with a light blue sky

    fill_ellipse(COLOR_YELLOW, -150, -150, 300, 300); // Sun

    fill_ellipse(COLOR_GREEN, 0, 400, 800, 400); // Hill

    fill_rectangle(COLOR_GRAY, 300, 300, 200, 200); // Base of house

    fill_rectangle(COLOR_LIGHT_BLUE, 400, 350, 50, 50); // Window

    draw_rectangle(COLOR_BLACK, 400, 350, 50, 50); // Window frame

    fill_triangle(COLOR_BROWN, 250, 300, 400, 150, 550, 300); // Roof of house

    fill_rectangle(COLOR_BROWN, 320, 400, 50, 100); // Door

    fill_ellipse(COLOR_BLACK, 320, 450, 10, 10); // Doorknob

    refresh_screen();

    delay(5000);
}
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
$ clang++ house.cpp -l SplashKit -o house.exe -Wall && ./house.exe

Atlas@DESKTOP-L0EPIQV MINGW64 /c/Users/Atlas/Desktop/code/p3
$ clang++ house.cpp -l SplashKit -o house.exe -Wall && ./house.exe

Atlas@DESKTOP-L0EPIQV MINGW64 /c/Users/Atlas/Desktop/code/p3
$ clang++ house.cpp -l SplashKit -o house.exe -Wall && ./house.exe

Atlas@DESKTOP-L0EPIQV MINGW64 /c/Users/Atlas/Desktop/code/p3
```

House drawing by Martin!

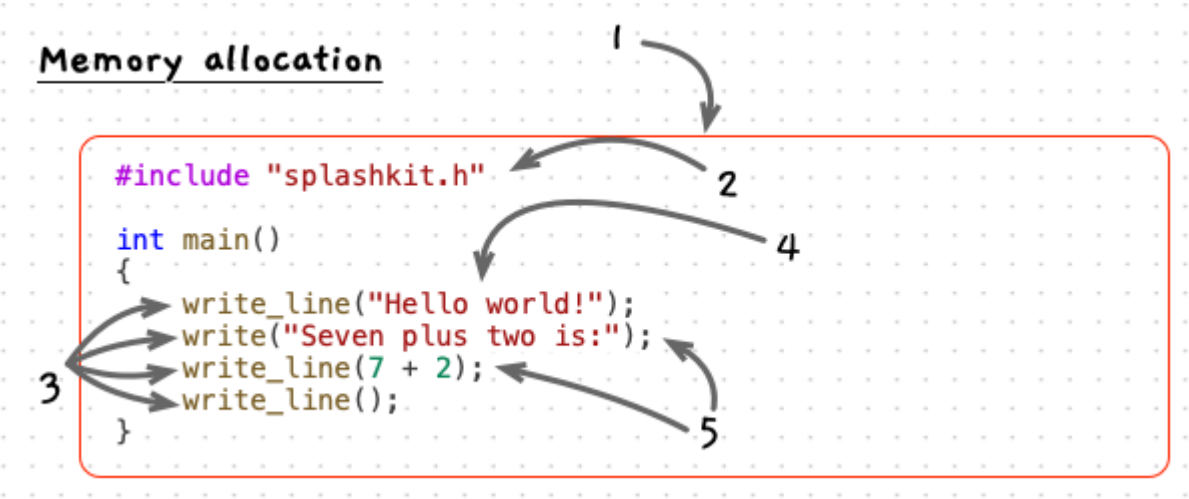
Group Activity

Optional – Group Activity
<provide a screenshot/photo of some material from the group activity. Did you collaborate via a google doc? Draw on the table? Show us that>
<The group activities are a good chance to work through the material with others and get - or offer - different perspectives. Did that help you? Tell us about it>

Brief Summary of Concepts

Concept	Key Idea / Description
Program	A list of instructions that a computer executes in a sequence, in order to reach a particular goal.

Statement	An authoritative command or action for a computer to perform as part of a larger sequence.
Procedure	A list of instructions that can be reused to achieve a more complex goal than can be done with a single statement, often with differing data passed in as arguments.
Procedure Call	A statement that instructs the computer to execute a procedure.
Argument	Data given to a procedure at runtime, allowing it to be executed in different contexts and achieve the same goal with different parameters
Expression	Code that represents a value or some kind of data
Type	The type of data stored in an expression, such as text, integer, boolean, float



Action	Response
1: Relate this code to the concepts of a program and sequence	This code is a sequence of commands carried out to achieve a goal
2: What is being done here? What does it give us access to?	This code calculates 7+2.
3: What are all these lines examples of?	Procedures with expressions passed in as arguments
4: We have several different things in the parentheses on these lines - what's the best overall term for "the thing in parentheses"?	Expressions
5: What might the difference between "write" and "write_line" be?	write_line likely outputs a new line after writing the expression passed in.
5 again: The things in parentheses here are different, but also similar. What will the `write_line` actually write to the terminal? How might you describe what is happening there using the terminology from this section?	The write_line(7 + 2); line will write the number 9 to the terminal, as there are no quotations around the argument of 7 + 2, meaning its type is not string, but integer.

house.cpp test-your-knowledge.cpp U

p3 > morse > test-your-knowledge.cpp > ...

```cpp
#include "splashkit.h"

int main()
{
    load_sound_effect("dit", "morse_dit.ogg");
    load_sound_effect("dah", "morse_dah.ogg");

    play_sound_effect("dit");
    delay(200);
    play_sound_effect("dit");
    delay(200);
    play_sound_effect("dit");
    delay(200);
    play_sound_effect("dit");
    delay(400);

    play_sound_effect("dit");
    delay(400);

    play_sound_effect("dit");
    delay(200);
    play_sound_effect("dah");
    delay(400);
    play_sound_effect("dit");
    delay(200);
    play_sound_effect("dit");
    delay(400);

    play_sound_effect("dit");
    delay(200);
    play_sound_effect("dah");
    delay(400);
    play_sound_effect("dit");
    delay(200);
    play_sound_effect("dit");
    delay(400);

    play_sound_effect("dit");
    delay(200);
    play_sound_effect("dit");
    delay(200);
    play_sound_effect("dit");
    delay(200);
}
```

Procedure calls: load_sound effect, play_sound_effect, delay

Arguments, expressions and literals: “dit”, “morse_dit.ogg”, “dah”, “morse_dah.ogg”, 200

This program demonstrates the concept of sequence by executing the commands in order, which plays a sequence of sounds to create a morse code message.

Reflection

What is the most important thing you learned from this and why?

I learnt how to use SplashKit's graphical and sound elements. This is more or less the only thing I learnt due to already having a good understanding of the concept of sequence in programming.

To enable you to better reflect on your performance over the long term in this unit, fill in the following google form: https://docs.google.com/forms/d/e/1FAIpQLSeaOSw_Haag9sgNX3qdrdZDMiY1wQoON9S5tqxD31UywFo8Vg/viewform

Form completed!

References

Generative AI acknowledgement

No generative AI was used.